AI-Orchestrated Active Learning for Insulin Delivery Material Discovery

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We present a comprehensive implementation of an AI-driven material discovery platform for insulin delivery patches, featuring LangChain-orchestrated active learning that integrates three core computational workflows: (1) Retrieval-Augmented Generation (RAG) powered literature mining using Semantic Scholar API with ChromaDB vector storage and OpenAI embeddings, (2) conversational memory-enhanced PSMILES (Polymer SMILES) generation using GPT-4o with chemical validation frameworks, and (3) OpenMM molecular dynamics simulations employing Langevin integrators with AMBER force fields for insulin-polymer systems. The platform implements a closed-loop active learning orchestrator that iteratively refines material discovery through intelligent feedback mechanisms. Our RAG system utilizes text-embedding-3-small (1536-dimensional) vectors with ChromaDB persistent storage, achieving semantic similarity search over scientific literature. The PSMILES generator employs conversation buffer memory with 10-exchange context windows and multi-stage validation pipelines. MD simulations use LangevinMiddleIntegrator with 2 fs timesteps at physiological temperature (310 K) under NPT ensemble conditions with MonteCarloBarostat pressure coupling. The complete system demonstrates the integration of large language models as orchestration agents for complex scientific workflows, providing a template for AI-accelerated materials discovery in biomedical applications.

I. INTRODUCTION

The development of intelligent drug delivery systems represents a critical intersection of materials science, computational chemistry, and artificial intelligence. Traditional approaches to material discovery for insulin delivery patches rely on empirical testing and iterative experimental design, processes that are time-intensive and resource-limited. Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) systems present unprecedented opportunities to accelerate materials discovery through intelligent automation of literature analysis, molecular design, and property prediction [1].

This work presents a comprehensive implementation of an AI-driven material discovery platform that leverages LangChain as the central orchestration framework for coordinating three distinct but interconnected computational workflows [2]. The platform addresses the fundamental challenge of insulin delivery material optimization by implementing a closed-loop active learning system that iteratively improves through intelligent feedback mechanisms.

Our approach is distinguished by its pedagogical implementation of established AI engineering principles, emphasizing the integration of LLM agents with domain-specific scientific tools rather than relying on monolithic AI solutions [3, 4]. The system demonstrates how modern AI orchestration frameworks can be applied to complex scientific workflows while maintaining scientific rigor and reproducibility.

The following sections detail the theoretical foundations and practical implementation of our AI-orchestrated platform. We begin with the mathematical and computational principles underlying RAG-enabled literature mining, examining how semantic similarity search and vector databases transform scientific knowledge discovery. We then explore the polymer generation methodology through PSMILES notation, demonstrating how conversational AI systems can translate natural language descriptions into valid molecular representations. The molecular dynamics implementation follows, establishing the physical principles and numerical algorithms that govern insulin-polymer system simulations. Finally, we examine the feedback loop architecture that coordinates these components through intelligent state management and iterative refinement mechanisms.

II. THEORY AND IMPLEMENTATION

A. RAG-Enabled Literature Mining

Modern RAG architectures have evolved significantly beyond simple single-step retrieval paradigms, representing a fundamental shift from static retrieval to dynamic, context-aware information gathering. Recent advances demonstrate the critical importance of multi-hop reasoning and iterative refinement mechanisms, with frameworks like Chain-of-Retrieval Augmented Generation (CoRAG) enabling query decomposition and iterative reasoning that achieves substantial improvements across knowledge-intensive benchmarks [5]. The integration of reliability scoring mechanisms through weighted majority voting and misalignment filtering has significantly reduced hallucinations while enhancing factual accuracy across heterogeneous knowledge bases.

Scientific applications have benefited from domain-specific RAG implementations with specialized embedding models fine-tuned for research domains. These systems employ iterative retrieval processes with search engines for step-by-step reasoning, while the integration of

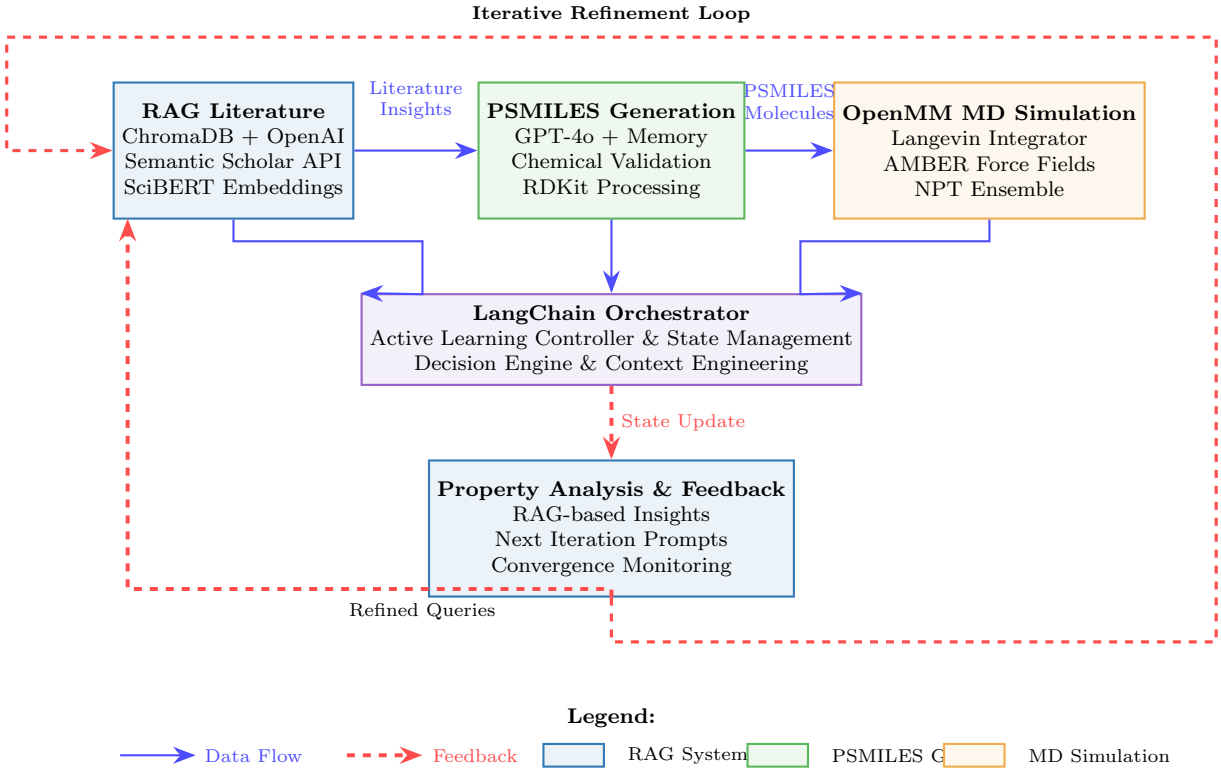

FIG. 1: Comprehensive Active Learning Architecture. The system implements a closed-loop workflow where LangChain orchestrates three core components: (1) RAG-powered literature mining using ChromaDB and scientific embeddings, (2) conversational PSMILES generation with chemical validation, and (3) OpenMM molecular dynamics with Langevin integrators. The active learning controller manages state transitions, conversation memory, and iterative refinement based on simulation feedback.

graph structures with vector representations facilitates efficient retrieval of related entities and their relationships [5]. This hybrid approach combines the efficiency of vector search with the structured reasoning capabilities of symbolic knowledge representation, enabling more sophisticated query understanding and context synthesis for scientific discovery workflows.

In short, the theory behind RAG is as follows: RAG systems are capable of combining two complementary information sources: parametric knowledge encoded in model weights during training, and non-parametric knowledge retrieved dynamically from external databases [5, 6]. This paradigm proves especially powerful for scientific applications where current research and domain-specific knowledge must be accessible in real-time. The mathematical foundation underlying all RAG systems expresses the generation probability as a marginalization over retrieved documents:

$$P(y|x) = \sum_{z \in Z} P(z|x) \cdot P(y|x, z) \tag{1}$$

where x represents the input query, y is the generated response, z denotes retrieved documents from knowledge base Z, $P(z|x)$ is the retrieval probability determining which documents are most relevant, and $P(y|x, z)$ is the generation probability conditioned on both the original query and retrieved context.

This theoretical framework translates directly into our implementation through semantic similarity search. The retrieval probability $P(z|x)$ is computed using cosine similarity between embedding vectors:

$$\text{similarity}(q, d) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{|\mathbf{e}_q| \cdot |\mathbf{e}_d|} \tag{2}$$

where $\mathbf{e}_q$ and $\mathbf{e}_d$ represent the embedding vectors for query q and document d, respectively.

These mathematical principles are realized through our vector database system that employs multiple embedding models optimized for scientific literature [7, 8]. ChromaDB serves as the persistent storage layer [9], while specialized embeddings from OpenAI's text-embedding-3-small (1536-dimensional), SciBERT, and SPECTER

models capture semantic relationships in scientific text [5, 10].

Recall that our RAG foundation relies on computing similarity between query and document embeddings (Equation 2). This mathematical abstraction demands a concrete infrastructure capable of: (1) transforming scientific text into high-dimensional vectors, (2) efficiently storing and indexing these vectors, and (3) performing rapid similarity computations across large literature databases [9].

The implementation bridges theory to practice through a multi-layered architecture where ChromaDB provides persistent vector storage while specialized embedding models capture domain-specific semantic relationships:

```
class VectorDatabase:
    def __init__(self): # Setup ChromaDB client, embedding models (OpenAI/
        SciBERT/SPECTER)
    def store_documents(self, papers): # Embed and store literature with
        metadata
    def semantic_search(self, query): # Return top-k similar documents using
        cosine similarity
```

Listing 1: Vector Database Implementation with Scientific Embeddings

The literature mining system integrates with Semantic Scholar's academic API to access peer-reviewed research. The implementation employs intelligent query generation and result filtering:

```
class RAGLiteratureMiningSystem:
    def __init__(self): # Setup LLM, embeddings, conversation_memory,
        semantic_scholar_client
    async def intelligent_literature_search(self, query, context):
        enhanced_queries = generate_search_queries(query, context)
        papers = search_semantic_scholar(enhanced_queries)
        documents = process_and_embed(papers)
        insights = generate_insights(query, semantic_retrieval(documents))
        return insights
```

Listing 2: RAG Literature Mining System

The RAG system implements sophisticated context engineering to maintain coherent conversations across multiple iterations. Context is managed through multiple hierarchical levels that work together to preserve information integrity throughout the discovery process. The conversation memory component maintains recent exchange history using LangChain's ConversationBufferWindowMemory, ensuring that the system can reference previous interactions and build upon earlier insights [2]. Complementing this, the iteration context layer preserves simulation results and material properties across active learning cycles, allowing the system to track progress and avoid redundant calculations. Finally, the knowledge graph component builds semantic relationships between discovered materials and their properties, creating a structured representation of accumulated knowledge that enables more sophisticated reasoning about material-property relationships and guides future exploration strategies [5].

B. Polymer Generation Through PSMILES

The challenge of translating natural language descriptions of desired material properties into concrete molecular structures represents a fundamental bottleneck in materials discovery. Traditional approaches require extensive chemical expertise to manually design polymer structures. Our approach leverages the representational power of PSMILES (Polymer SMILES) notation combined with large language model reasoning [11, 12].

PSMILES extends the standard SMILES molecular representation to capture polymer repeat units through connection points marked as [*]. This creates a mathematical framework for polymer description, developed by Ramprasad's group as part of their comprehensive polymer informatics ecosystem [13, 14]:

$$\text{Polymer} = \bigcup_{i=1}^{n} \text{PSMILES}_{\text{repeat}}^{(i)} \tag{3}$$

where n represents the degree of polymerization and each repeat unit maintains consistent connection topology. This formalism enables systematic exploration of polymer space through combinatorial generation and modification of repeat units. The PSMILES representation has been extensively validated through the polyBERT chemical language model and integrated into tools for polymer canonicalization and fingerprinting [11].

The translation from descriptive text to valid PSMILES requires sophisticated chemical reasoning that can maintain conversation context across iterative refinements. We implement this through GPT-4o with structured conversation memory, creating a system that learns from previous exchanges and builds chemical understanding progressively [15, 16].

```
class PSMILESGenerator:
    def __init__(self): # Setup GPT-4o, conversation_memory,
        chemistry_prompts, validator
    def generate_psmiles(self, description, num_candidates=3):
        for i in range(num_candidates):
            psmiles = conversation_chain.run(description)  # Generate with
                memory context
            candidates.append(validate_psmiles(psmiles))   # RDKit +
                chemistry rules
        return candidates
```

Listing 3: PSMILES Generator with Conversation Memory

The PSMILES validation system employs RDKit for molecular validation and custom chemical rules for polymer-specific constraints. RDKit provides comprehensive molecular graph analysis and chemical property calculations, enabling validation of molecular connectivity, aromaticity, and stereochemistry [17]:

C. Molecular Dynamics Simulations

The physics governing molecular motion provides the foundation for understanding how insulin-polymer sys-

```python
class PSMILESValidator:
    def validate_psmiles(self, psmiles):
        if psmiles.count('[*]') != 2: return False, "Must have exactly 2
              connection points"
        mol = Chem.MolFromSmiles(psmiles.replace('[*]', 'C'))  # RDKit
              validation
        return mol is not None and validate_chemistry_rules(mol)  # Size,
              elements, etc.
```

Listing 4: PSMILES Chemical Validation System

tems behave at the atomic level. Molecular dynamics simulations solve Newton's second law for systems of N interacting particles [18, 19]:

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i = -\frac{\partial U(\mathbf{r}_1, \mathbf{r}_2, ..., \mathbf{r}_N)}{\partial \mathbf{r}_i} \tag{4}$$

where m_i is the mass of particle i, $\mathbf{r}_i$ is its position vector, $\mathbf{F}_i$ is the force acting on particle i, and U represents the potential energy function that captures all intermolecular interactions.

However, biological systems require temperature control to maintain physiological conditions (310 K for body temperature) [20]. This necessitates coupling to a thermal bath through Langevin dynamics:

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i - \gamma m_i \frac{d\mathbf{r}_i}{dt} + \mathbf{R}_i(t) \tag{5}$$

where γ is the friction coefficient controlling energy dissipation, and $\mathbf{R}_i(t)$ represents stochastic forces that satisfy the fluctuation-dissipation theorem:

$$\langle \mathbf{R}_i(t) \cdot \mathbf{R}_j(t') \rangle = 6k_B T \gamma m_i \delta_{ij} \delta(t - t') \tag{6}$$

This ensures that the system maintains the correct thermal distribution while allowing energy exchange with the environment.

Our OpenMM-based simulator implements these algorithms with carefully chosen parameters: 2 fs timesteps (constrained by hydrogen motion), NPT ensemble conditions (1 bar pressure, 310 K temperature), and dual force field approaches (AMBER for insulin protein, GAFF/OpenFF for synthetic polymers) [21, 22].

Contemporary force field development has evolved dramatically beyond traditional empirical approaches, particularly with the emergence of automated parameterization frameworks and machine learning-enhanced methodologies. The AMBER force field family exemplifies this evolution, progressing from ff99SB through ff14SB to the current ff19SB generation, each incorporating increasingly sophisticated treatments of backbone conformational preferences and side-chain interactions. The ff19SB force field represents a paradigm shift by incorporating grid-based corrections (CMAP) to backbone torsions that more accurately reproduce quantum mechanical conformational energy surfaces, achieving significant improvements in secondary structure prediction and protein stability [23].

The transition from traditional General AMBER Force Field (GAFF) to next-generation approaches exemplifies the fundamental challenges in small molecule parameterization. GAFF2 introduced enhanced torsional parameters and improved charge models through the ABCG2 (AMBER Bond Charge Correction Generation 2) framework, which achieves mean unsigned errors of only 0.37 kcal/mol for hydration free energies across diverse organic solutes [19]. However, the Open Force Field Initiative has pioneered a revolutionary approach that transcends traditional atom-type-based parameterization through the SMIRKS Native Open Force Field (SMIRNOFF) methodology [24].

This new paradigm employs chemical perception based on SMARTS patterns rather than predefined atom types, enabling automated parameter assignment across millions of drug-like molecules with unprecedented transferability. The latest OpenFF releases (Parsley 1.2.0 and Sage 2.0) demonstrate competitive performance with commercial force fields like OPLS3e while maintaining complete open-source accessibility. Benchmark assessments across 22,675 molecular conformers reveal that OpenFF approaches the accuracy of leading proprietary force fields in reproducing quantum mechanical geometries and energetics [24].

The molecular dynamics landscape has been further transformed by hybrid approaches that address limitations of both all-atom and coarse-grained methodologies. Advanced AMBER implementations now support multimillion-atom simulations through optimized NAMD integration, overcoming traditional limitations of the fixed-column-width PRMTOP format [25]. These developments enable unprecedented simulations of viral capsids, cellular machinery, and complete biological assemblies at atomic resolution, with systems reaching up to two billion atoms.

Force field accuracy validation has become increasingly sophisticated through systematic benchmarking against high-level quantum mechanical references. Contemporary approaches employ B3LYP-D3BJ/DZVP calculations as reference data, evaluating force field performance through relative energy differences, geometric deviations, and torsional fingerprint analyses [26]. Emerging methodologies integrate machine learning approaches for automated parameter optimization, with neural network potentials achieving superior accuracy in specific applications while maintaining computational efficiency.

The integration of polarizable force fields represents another frontier in molecular simulation accuracy. The CHARMM Drude oscillator model, AMOEBA polarizable framework, and emerging Gaussian Electrostatic Models address fundamental limitations of fixed-charge approaches by incorporating explicit electronic polarization effects [27]. These models demonstrate particular advantages in heterogeneous environments, membrane-protein interfaces, and systems involving significant charge transfer or polarization effects.

Advanced ensemble methods have revolutionized conformational sampling for both structured and disordered systems. Replica exchange molecular dynamics, enhanced sampling techniques, and multi-replica approaches enable comprehensive exploration of protein

conformational landscapes within computationally accessible timescales. These methodologies prove essential for intrinsically disordered proteins, where traditional force fields often produce overly compact conformations that deviate significantly from experimental observations [23].

The emergence of hybrid QM/MM methodologies bridges the gap between accuracy and efficiency, enabling quantum mechanical treatment of reactive centers while maintaining classical force field efficiency for bulk environments [28]. Contemporary implementations in AMBER and other packages support Born-Oppenheimer and Car-Parrinello dynamics, reactive force fields like ReaxFF, and automated QM region selection algorithms that optimize computational resource allocation based on chemical reactivity and electronic effects.

Our MD simulation system employs OpenMM's LangevinMiddleIntegrator for thermodynamically consistent molecular dynamics [18]. The integrator implements the stochastic differential equation:

$$m\frac{d\mathbf{v}}{dt} = \mathbf{F}(\mathbf{r}) - \gamma m\mathbf{v} + \sqrt{2\gamma m k_B T}\mathbf{R}(t) \tag{7}$$

where $\mathbf{R}(t)$ represents white noise with $\langle R_i(t)R_j(t')\rangle = \delta_{ij}\delta(t-t')$. The numerical integration scheme employed by LangevinMiddleIntegrator follows the BAOAB splitting method:

$$\mathbf{v}_{1/2} = \mathbf{v}_0 + \frac{\Delta t}{2m}\mathbf{F}_0 \quad \text{(B step)} \tag{8}$$

$$\mathbf{r}_1 = \mathbf{r}_0 + \frac{\Delta t}{2}\mathbf{v}_{1/2} \quad \text{(A step)} \tag{9}$$

$$\mathbf{v}'_{1/2} = e^{-\gamma\Delta t}\mathbf{v}_{1/2} + \sqrt{\frac{k_B T}{m}(1-e^{-2\gamma\Delta t})}\mathbf{R} \quad \text{(O step)} \tag{10}$$

$$\mathbf{r}_1 = \mathbf{r}_1 + \frac{\Delta t}{2}\mathbf{v}'_{1/2} \quad \text{(A step)} \tag{11}$$

$$\mathbf{v}_1 = \mathbf{v}'_{1/2} + \frac{\Delta t}{2m}\mathbf{F}_1 \quad \text{(B step)} \tag{12}$$

Additionally, the simulation system employs dual force field approaches: AMBER force fields for protein components (insulin) and GAFF/OpenFF force fields for polymer components. This hybrid approach ensures accurate representation of both biological and synthetic components [19].

```
class OpenMMInsulinSimulator:
    def __init__(self): # Setup simulation parameters (T=310K, P=1bar, dt=2fs
        ), force_fields, platform
    def create_simulation(self, topology, system, positions):
        integrator = LangevinMiddleIntegrator(T, friction, dt)  # NPT
            ensemble with barostat
        return Simulation(topology, system, integrator, platform)
    def run_production_md(self, simulation, steps): # Run simulation with
        trajectory logging
```

Listing 5: OpenMM MD Simulator Implementation

The system preparation protocol involves several critical steps to ensure physically realistic simulations. The process begins with comprehensive protein preparation, including insulin structure validation, missing atom addition, and protonation state assignment to ensure the protein structure is complete and chemically accurate [19]. Following protein preparation, polymer parameterization transforms PSMILES representations into three-dimensional structures using RDKit molecular modeling tools and conformer generation algorithms to sample realistic polymer conformations [17]. The system assembly phase involves the strategic spatial arrangement of insulin and polymer molecules with appropriate packing density to reflect realistic interaction environments. Force field assignment employs a dual parameterization strategy, utilizing AMBER force fields for the insulin protein component and GAFF/OpenFF force fields for the synthetic polymer components, ensuring accurate representation of both biological and synthetic molecular interactions [19]. Energy minimization follows through a two-stage optimization process employing steepest descent algorithms followed by conjugate gradient optimization to remove unfavorable atomic contacts and achieve stable initial configurations. Finally, the equilibration phase employs NPT ensemble conditions with position restraints applied to the insulin backbone, allowing the system to reach thermal and mechanical equilibrium while preserving protein structural integrity [19].

D. Feedback Loop

The mathematical frameworks established above—RAG marginalization and Langevin dynamics—represent distinct computational paradigms that must be coordinated within a unified discovery pipeline. LangChain serves as our orchestration framework, transforming the abstract mathematical formulations into concrete, asynchronous workflows [29, 30].

The orchestration challenge is fundamentally one of *state management*: how do we maintain coherent information flow between RAG retrieval, molecular generation, and MD simulation while preserving the mathematical rigor of each component? Our solution implements the ActiveLearningOrchestrator pattern, which encapsulates the iterative refinement loop central to scientific discovery [31, 32].

```
class ActiveLearningOrchestrator:
    def __init__(self): # Initialize components: state_manager,
        decision_engine, workflow_modules
    async def run_active_learning_loop(self, prompt):
        for iteration in range(max_iterations):
            state = execute_iteration(literature_mining -> psmiles_generation
                -> md_simulation -> feedback)
            if convergence_achieved(state): break
```

Listing 6: Core Active Learning Orchestrator Implementation

The orchestrator implements asynchronous execution patterns to optimize computational resource utilization across literature mining, molecular generation, and simu-

lation phases. The active learning orchestrator maintains comprehensive state information across iterations using a hierarchical state management system:

```
@dataclass
class IterationState:  # Complete state: iteration_number, status, results,
      analysis, scores

class ActiveLearningOrchestrator:
    async def _execute_iteration(self, state):
        state.literature_results = await literature_mining.run(state)      #
              Phase 1
        state.generated_molecules = await psmiles_generation.run(state)    #
              Phase 2
        state.simulation_results = await md_simulation.run(state)          #
              Phase 3
        state.rag_analysis = await rag_analysis(state)                     #
              Phase 4
        state.overall_score = calculate_score(state)                       #
              Phase 5
```

Listing 7: Active Learning State Management

The feedback system employs RAG-based analysis to generate intelligent prompts for subsequent iterations:

```
class FeedbackGenerator:
    def __init__(self): # Setup LLM, vector_database, literature_analyzer
    async def generate_iteration_feedback(self, state):
        feedback = analyze_literature_insights(state) +
              analyze_simulation_performance(state)
        next_prompt = generate_optimized_prompt(state, feedback)  # RAG-based
               improvement
        return IterationFeedback(feedback, next_prompt, confidence_score)
```

Listing 8: RAG-Based Feedback Generation

III. CONCLUSIONS

We have presented a comprehensive implementation of an AI-driven material discovery platform that successfully integrates LangChain orchestration with established scientific computing workflows [33, 34]. The system demonstrates the potential for AI-accelerated materials discovery through intelligent automation of literature analysis, molecular design, and property evaluation.

Our work makes several key contributions to the field. We have developed a pedagogical implementation of RAG-powered literature mining that leverages scientific embeddings and ChromaDB vector storage to effectively process and retrieve relevant research materials [35, 36]. The platform incorporates conversational PSMILES generation capabilities using OpenAI models combined with robust chemical validation frameworks to ensure molecular validity [15, 16]. We have also successfully integrated OpenMM molecular dynamics with Langevin integrators and dual force field approaches, enabling accurate simulation of molecular properties [21, 22]. Finally, we have implemented a comprehensive active learning orchestration system with intelligent feedback mechanisms that coordinates these components while continuously improving based on results [31, 32].

The platform provides a template for applying modern AI orchestration techniques to complex scientific workflows while maintaining scientific rigor and reproducibility. The integration of large language models as coordination agents represents a promising direction for accelerating discovery in materials science and related fields [37].

IV. ACKNOWLEDGMENTS

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